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Computationally efficient spatio-temporal disease mapping for big data

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ABSTRACT

Disease mapping models estimate the spatio-temporal variation in population-level disease risks or rates across a set of K areal units for N time periods, aiming to identify temporal trends and spatial hotspots. Highly parameterised Bayesian hierarchical models with over KN random effects are commonly used to estimate this spatio-temporal variation, which are assigned autoregressive and conditional autoregressive prior distributions. These models work well when there are tens of thousands of data points, but are likely to be computationally burdensome when this rises to hundreds of thousands or above. This paper proposes a computationally efficient alternative, which can fit a range of spatio-temporal disease trends almost as well as existing highly parameterised models but only takes around 5% to 40% of the time to implement. It achieves this by modelling the average spatial and temporal trends in the data with autoregressive type random effects, which are augmented by an observation-driven process using functions of earlier data as additional covariates in the model. The efficacy of this methodology is tested by simulation, before being applied to the motivating study that estimates the spatio-temporal trends in asthma, cancer, coronary heart and chronic obstructive pulmonary disease prevalences for $K = 32,751$ small areas over $N = 13$ years in England.

1. Introduction

The prevalence of disease varies in space and time, and the field of disease mapping makes inference about this spatio-temporal variation using population-level summary data relating to a set of K non-overlapping areal units for N consecutive time periods. This inference allows public health professionals to answer a number of important policy questions, including: (i) what risk factors are associated with increased disease rates (e.g., Djeudeu et al., 2020); (ii) how big are the spatial inequalities in disease rates that measure the inequity of society (e.g., Rahimzadeh et al., 2021); and (iii) which areas exhibit the highest disease rates and/or increasing trends over time that should be the focus for interventions (e.g., Anderson et al., 2017). Disease mapping is a fast growing research field encompassing computing science, epidemiology, geography and statistics, and a recent review is given by MacNab (2022).

Bayesian hierarchical models with spatio-temporally autocorrelated random effects are the predominant approach for modelling these data, which stems from Tobler's first law of geography that states *'everything is related to everything else, but near things are more related than distant things'*. This implies that geographically and temporally neighbouring observations are likely to have similar (correlated) disease rates, and these random effects models borrow strength in the estimation by smoothing the disease rate data over space and time. Gaussian Markov random field (GMRF, Rue and Held, 2005) priors are the most popular way to enforce this smoothness, including autoregressive processes in time and conditional autoregressive (CAR, Besag et al., 1991) processes in

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space. The popularity of these models is due to their computational efficiency (i.e., they have sparse precision matrices), flexibility (i.e., they are non-stationary and anisotropic), and ease of implementation in R.

A range of GMRF-based models have been proposed, including general-purpose smoothing models as well as specialised models designed for tasks such as cluster detection (e.g., Jaya and Folmer, 2021) and boundary identification (e.g., Lee et al., 2021). The general-purpose models are the focus here, and the simplest is the separable structure proposed by Knorr-Held and Besag (1998) that is computationally efficient to fit but makes the restrictive assumption that the spatial disease rate surface does not change over time. This model was extended by Knorr-Held (2000) who augmented it with a set of spatio-temporal interactions. These interactions are assigned one of four prior distributions, allowing them to be either independent or correlated in space, time or both space and time. This structure remains one of the most commonly used today, while the other popular approach is the temporally autoregressive process model with spatially autocorrelated errors presented in Rushworth et al. (2014).

The choice of spatio-temporal smoothing model requires the user to trade-off flexibility against computational overhead, and for most applications computational time is not a major concern because of the relatively small data sizes (up to a few thousand areal units). However, an emerging research question is how to scale up these models for analysing *big data*, where the numbers of areal units are in the tens of thousands. Orozco-Acosta et al. (2021, 2023) propose a spatial *divide and conquer* approach, which fits separate models to different spatial subsets of the data before combining the resulting posterior disease rate distributions. However, their approach requires the user to partition the study region into smaller sub-units, and the impact of this choice of sub-units on model inference has not been well studied.

Therefore, this paper proposes a novel spatio-temporal smoothing model for big data called STARPOD (Spatio-Temporal AutoRegressive Parameter and Observation Driven model), which has the ability to accurately capture a range of spatio-temporal disease rate structures whilst being computationally efficient to implement for big data. It achieves this by modelling the separable spatio-temporal structure in the data with traditional GMRF priors, whilst augmenting this with an observation-driven autoregressive process (Zeger and Qaqish, 1988) using functions of earlier data as additional covariates to capture any changing spatial dynamics over time. The efficacy of our proposed model is illustrated by simulation studies in Section 5, which show that it has a similar estimation performance to the commonly used complex models of Knorr-Held (2000) and Rushworth et al. (2014), but at a computational overhead that is only slightly more than the simple (and fast) separable model of Knorr-Held and Besag (1998). The development of this methodology is motivated by a new study aiming to quantify the spatio-temporal dynamics of four of the most prevalent diseases in England, namely asthma, cancer, coronary heart disease and chronic obstructive pulmonary disease. A summary of the data and the motivating questions are presented in Section 2, while the results of the study are presented in Section 6. Before then a review of spatio-temporal disease mapping is given in Section 3, while our methodological contribution is outlined in Section 4. Finally, Section 7 concludes the paper summarising the key findings and areas for future work.

2. Motivating study

2.1. Study data

The study region of mainland England is partitioned into $K = 32{,}751$ Lower Super Output Areas (LSOA), which in 2017 had a median population of 1687, a 1st percentile of 1018 and a 99th percentile of 3248. Yearly data summarising the prevalences of 4 common diseases for the $N = 13$ year period between 2005 and 2017 were obtained for this set of LSOAs from the Place-Based Longitudinal Data Resource (PLDR, https://pldr.org/). These diseases are: (i) asthma; (ii) cancer (excluding non-melanotic skin cancers); (iii) coronary heart disease (CHD); and (iv) chronic obstructive pulmonary (COPD); which were chosen because they are four of the leading causes of chronic ill health in the UK. The prevalence of each disease summarises the proportion of people in each LSOA and year that are living with that disease, provided they are registered with a general practice (doctors) surgery. Note, one LSOA (Kensington and Chelsea 018C) was removed because the yearly prevalence data appeared to be implausible, as for example the prevalence for cancer jumped from 1.4% of the population in 2010 to 6.8% of the population in 2011. Disease prevalence for each LSOA in each year is quantified by the number of people living with disease divided by the total number of people. The temporal trends in these raw prevalences for each disease are displayed in Fig. 1 by boxplots (presented on the percentage scale). The figure shows that neither asthma nor CHD exhibit any prominent long-term trends, with average changes over the 13-year study period being only 0.2% for asthma and 0.3% for CHD. In contrast, the median cancer prevalence has increased from 0.5% in 2005 to 2.7% in 2017, while COPD prevalence has increased from 1.3% to 1.9% over the same time period. The figure also shows that asthma is the most prevalent of the four diseases with a median prevalence of 6.0%, while cancer is the least prevalent with a median of 1.5%. Additional exploratory analyses focusing on their spatial trends and between disease correlations are presented in Section 1.1 of the supplementary material.

2.2. Aims and motivating questions

The methodological aim of this paper is to develop a computationally efficient model for estimating non-separable spatio-temporal disease trends in big data, which includes the motivating study that has 425,763 data points for each disease. Within this study the data analysis aims to answer the following important public health questions.

(A). What are the overall temporal trends in disease prevalence for each disease, and which of the 9 regions of England exhibit the highest and lowest prevalences?

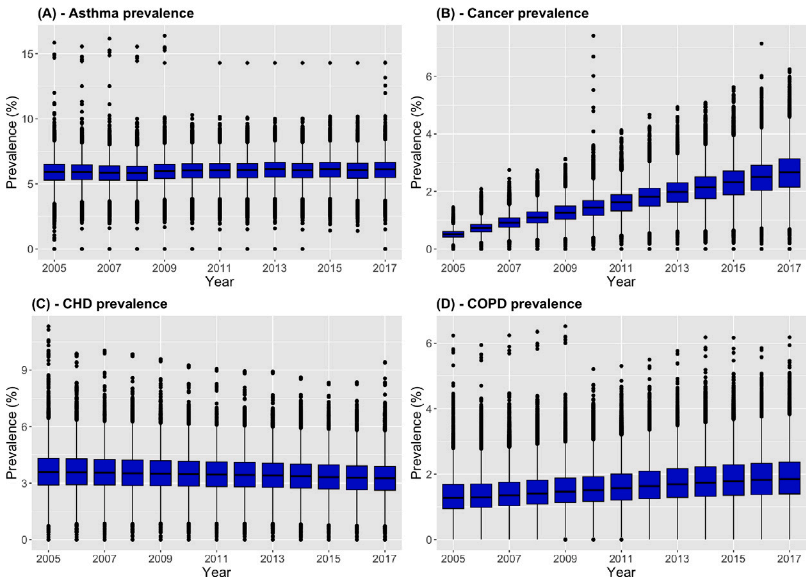

Fig. 1. Boxplots showing the temporal trends in the noisy raw prevalences (as a %) of: (A) Asthma; (B) Cancer; (C) Coronary Heart Disease; and (D) Chronic Obstructive Pulmonary Disease.

(B). How big are the: (i) total; and (ii) socio-economic; health inequalities in the prevalences of the four diseases across England, and how do these vary regionally and over time?

(C). Which LSOAs have the highest and lowest disease prevalences in England for each disease?

For the purposes of the above questions England has been partitioned into 9 large regions, which are often used when comparing the health and wealth across different parts of the country. A map showing the geographical extent of these regions is presented in Section 1.2 of the supplementary material. Answering questions (A) and (C) allows policymakers to identify which regions and local communities exhibit the highest disease burden, as well as which diseases are increasing in prevalence and hence potentially need mitigating interventions. In contrast, answering question (B) allows us to quantify the level of inequality in disease prevalence across England, which is a barometer of the fairness of society (Public Health England, 2021). Total inequality measures the overall variation in disease prevalence between different communities, and is measured here by the standard deviation in the estimated disease prevalences between LSOAs. Additionally, socio-economic inequality measures the differences in disease prevalence between rich and poor communities, and the level of socio-economic deprivation in each LSOA is measured here by the English index of multiple deprivation (IMD).

3. Spatio-temporal disease mapping

The study region is partitioned into K non-overlapping areal units $\mathbb{S} = \{\mathcal{A}_1, \ldots, \mathcal{A}_K\}$, and data on disease incidence/prevalence in each areal unit is available for $t = 1, \ldots, N$ consecutive time periods. The data for areal unit $\mathcal{A}_k$ and time period t include $\{Y_t(\mathcal{A}_k), O_t(\mathcal{A}_k)\}$, where $Y_t(\mathcal{A}_k)$ denotes the observed disease incidence/prevalence count while $O_t(\mathcal{A}_k)$ denotes a population offset. The latter controls for the varying population sizes and possibly demographic structures between the K areal units and N time periods, and in practice, $O_t(\mathcal{A}_k)$ represents either the total population size or an indirectly standardised expected disease count. A generalised linear mixed model is used to model the spatio-temporal variation in disease rates, with inference commonly set in a Bayesian hierarchical framework. Both Markov chain Monte Carlo (MCMC) simulation (e.g., using the R package `CARBayesST`, Lee et al., 2018) and integrated nested Laplace approximations with variational Bayes approximations (e.g., using the R package `INLA`, Rue et al., 2009) have been used to estimate the model parameters, with the latter being preferred for big data due to its superior computational speed. The general model form is given by

$$\begin{aligned} Y_t(\mathcal{A}_k) &\sim f\{Y_t(\mathcal{A}_k)|\mu_t(\mathcal{A}_k), O_t(\mathcal{A}_k)\} \\ g\{\mu_t(\mathcal{A}_k)\} &= \beta_0 + \xi_t(\mathcal{A}_k), \end{aligned} \tag{1}$$

where $\mu_t(\mathcal{A}_k) = \mathbb{E}\{Y_t(\mathcal{A}_k)\}$, $g(.)$ is an invertible link function and β_0 is an intercept term. Covariates could be added to the above model, but unlike ecological regression applications most disease mapping studies do not include them (e.g., Knorr-Held, 2000 and Orozco-Acosta et al., 2021). The two commonly used special cases of this model are: (i) the Poisson log-linear specification $Y_t(\mathcal{A}_k) \sim \text{Poisson}\{O_t(\mathcal{A}_k)\theta_t(\mathcal{A}_k)\}$ and $\ln\{\theta_t(\mathcal{A}_k)\} = \beta_0 + \xi_t(\mathcal{A}_k)$, where $O_t(\mathcal{A}_k)$ is an indirectly standardised expected disease count; and (ii) the binomial logistic specification $Y_t(\mathcal{A}_k) \sim \text{Binomial}\{O_t(\mathcal{A}_k), \theta_t(\mathcal{A}_k)\}$ and $\ln\{[\theta_t(\mathcal{A}_k)]/[1 - \theta_t(\mathcal{A}_k)]\} = \beta_0 + \xi_t(\mathcal{A}_k)$, where

$O_t(\mathcal{A}_k)$ denotes the total population size. Spatio-temporal independence is assumed here at the data likelihood level, because the spatio-temporal trend and autocorrelation in the data are captured through $\boldsymbol{\xi} = \{\xi_t(\mathcal{A}_k)\}$ which often contains multiple sets of spatial and temporal random effects. Temporal random effects are often represented by autoregressive (AR) processes, while spatial random effects are captured by conditional autoregressive (CAR, Besag et al., 1991) processes. The latter utilise a binary $K \times K$ neighbourhood matrix $\mathbf{W}$, which determines the spatial closeness between the K areal units. Typically, the kjth element of this matrix $w_{kj} = 1$ if areal units $\{\mathcal{A}_k, \mathcal{A}_j\}$ share a common border and hence are spatially close, with $w_{kj} = 0$ otherwise and $w_{kk} = 0$ for all k. The most popular class of spatio-temporal disease mapping model was proposed by Knorr-Held (2000), who represents $\boldsymbol{\xi}$ by the following additive decomposition:

$$\xi_t(\mathcal{A}_k) = \phi(\mathcal{A}_k) + \nu(\mathcal{A}_k) + \gamma_t + \delta_t + \epsilon_t(\mathcal{A}_k). \tag{2}$$

Here, $\boldsymbol{\phi} = [\phi(\mathcal{A}_1), \ldots, \phi(\mathcal{A}_K)]$ and $\boldsymbol{\nu} = [\nu(\mathcal{A}_1), \ldots, \nu(\mathcal{A}_K)]$ are $K \times 1$ vectors of spatial random effects while $\boldsymbol{\gamma} = [\gamma_1, \ldots, \gamma_N]$ and $\boldsymbol{\delta} = [\delta_1, \ldots, \delta_N]$ are $N \times 1$ vectors of temporal random effects. Each one is assigned a GMRF prior with a different precision matrix, that is: $\boldsymbol{\phi} \sim \mathrm{N}(\mathbf{0}, \tau_\phi^{-1}\mathbf{K}_\phi^{-1})$, $\boldsymbol{\nu} \sim \mathrm{N}(\mathbf{0}, \tau_\nu^{-1}\mathbf{K}_\nu^{-1})$, $\boldsymbol{\gamma} \sim \mathrm{N}(\mathbf{0}, \tau_\gamma^{-1}\mathbf{K}_\gamma^{-1})$, $\boldsymbol{\delta} \sim \mathrm{N}(\mathbf{0}, \tau_\delta^{-1}\mathbf{K}_\delta^{-1})$, where $(\tau_\phi, \tau_\nu, \tau_\gamma, \tau_\delta)$ are the respective precision parameters. The time averaged spatial structure in the data is modelled by the sum of a spatially correlated process $\boldsymbol{\phi}$ and an independent process $\boldsymbol{\nu}$, the latter having $\mathbf{K}_\nu = \mathbf{I}$ where $\mathbf{I}$ denotes the identity matrix. The former is represented by an intrinsic CAR process with $\mathbf{K}_\phi = \mathrm{Diag}(\mathbf{W1}) - \mathbf{W}$, where $\mathrm{Diag}(\mathbf{W1})$ denotes a diagonal matrix containing the row sums of $\mathbf{W}$. The spatially averaged temporal trend in the data is also modelled by the sum of correlated ($\boldsymbol{\gamma}$) and independent ($\boldsymbol{\delta}$) processes, where the latter has $\mathbf{K}_\delta = \mathbf{I}$ while the former is modelled with a first-order random walk where $\mathbf{K}_\gamma = \mathbf{D}$ denotes a first-order difference matrix.

The inclusion of just $(\boldsymbol{\phi}, \boldsymbol{\nu}, \boldsymbol{\gamma}, \boldsymbol{\delta})$ in $\boldsymbol{\xi}$ produces a separable spatio-temporal structure, where the same spatial pattern is assumed for all time periods. This has the advantage of simplicity and parsimony as it only contains $2K + 2N$ random effects, but is restrictive because the spatial pattern in real data is likely to evolve over time. Any spatio-temporal interactions in disease rates are captured in (2) by $\boldsymbol{\epsilon} = [\epsilon_1(\mathcal{A}_1), \ldots, \epsilon_N(\mathcal{A}_K)]$, which are modelled as $\boldsymbol{\epsilon} \sim \mathrm{N}(\mathbf{0}, \tau_\epsilon^{-1}\mathbf{K}_\epsilon^{-1})$. Knorr-Held (2000) proposes modelling $\mathbf{K}_\epsilon$ as the Kronecker product of a temporal (either $\mathbf{K}_\gamma$ or $\mathbf{K}_\delta$) and a spatial (either $\mathbf{K}_\phi$ or $\mathbf{K}_\nu$) structure matrix, resulting in four possible models. The simplest assumes independence in space and time and has $\mathbf{K}_\epsilon = \mathbf{K}_\nu \otimes \mathbf{K}_\delta$, which is known as a Type-I interaction. Conversely, the most complex allows for correlation in space and time and has $\mathbf{K}_\epsilon = \mathbf{K}_\phi \otimes \mathbf{K}_\gamma$, which is known as a Type-IV interaction. Whichever interaction is chosen this model can capture a much richer set of spatio-temporal structures than the simpler separable alternative, but this comes at the cost of being much less computationally efficient to fit as it contains $2K + 2N + KN$ random effects rather than $2K + 2N$.

Another popular spatio-temporal smoothing model was proposed by Rushworth et al. (2014), which combines (1) with the multivariate first-order autoregressive process model for $\boldsymbol{\xi}_t = \{\xi_t(\mathcal{A}_1), \ldots, \xi_t(\mathcal{A}_K)\}$ given by

$$\begin{aligned} \boldsymbol{\xi}_t | \boldsymbol{\xi}_{t-1} &\sim \mathrm{N}(\alpha\boldsymbol{\xi}_{t-1}, \tau_\xi^{-1}\mathbf{K}_\xi) \quad \text{for } t = 2, \ldots, N \\ \boldsymbol{\xi}_1 &\sim \mathrm{N}(\mathbf{0}, \tau_\xi^{-1}\mathbf{K}_\xi). \end{aligned} \tag{3}$$

Here, the precision matrix $\mathbf{K}_\xi = \rho\{\mathrm{Diag}(\mathbf{W1}) - \mathbf{W}\} + (1-\rho)\mathbf{I}$ corresponds to the CAR prior proposed by Leroux et al. (2000), where (α, ρ) respectively control the level of temporal and spatial autocorrelation in the data. Values of zero for both parameters correspond to independence, while values of 1 correspond to strong positive dependence. Numerous other spatio-temporal disease mapping models have been proposed in the literature, with Bernardinelli et al. (1995) and Waller et al. (1997) being two of the most popular. However, the former forces the temporal trend in disease risk to be linear in each areal unit, while the latter assumes completely independent spatial surfaces for each time period, neither of which are likely to be realistic assumptions.

These Bayesian disease mapping models are completed by specifying hyperprior distributions, and here we choose weakly informative specifications to let the data speak for themselves. Specifically: intercept term - $\beta_0 \sim \mathrm{N}(0, 100000)$; precision parameters (with various subscripts) - $\ln(\tau^{-1/2}) \sim \text{Half-Normal}(\tau_0 = 0.001)$, a half normal prior on the standard deviation scale as discussed by Gelman (2006); dependence parameters - $\ln\left(\frac{\rho}{1-\rho}\right) \sim \mathrm{N}(0, 0.25)$ and $\ln\left(\frac{1+\alpha}{1-\alpha}\right) \sim \mathrm{N}(0, 0.25)$. Further details are given in Section 2.1 of the supplementary material.

4. Methodology

This section proposes a novel Spatio-Temporal AutoRegressive Parameter and Observation Driven (STARPOD) model for big-data disease mapping applications. The model is much more computationally efficient to fit than the complex models outlined in the previous section, and has sufficient flexibility to capture a range of spatio-temporal patterns in disease rates almost as well as these existing highly parameterised alternatives. The model is set within a Bayesian hierarchical framework and uses the `R` package `INLA` for parameter estimation, which matches the fitting approach used for the competitor models outlined above.

4.1. Level 1 - Data likelihood model

The data likelihood component has the general form (1), with Poisson log linear and binomial logistic models being the most popular special cases. The linear predictor is based on a combination of parameter and observation driven components, with the average spatial and temporal patterns being captured by random effects while local deviations from these average patterns are captured by functions of past data values. The latter are motivated by observation-driven models that date back to Zeger and Qaqish (1988) from the time series literature. In terms of notation, let $\mathbf{Y}_t = \{Y_t(\mathcal{A}_1), \ldots, Y_t(\mathcal{A}_K)\}$ and $\mathbf{O}_t = \{O_t(\mathcal{A}_1), \ldots, O_t(\mathcal{A}_K)\}$ represent the observed disease data and the population offsets for time period t, while $\mathbf{Y} = \{\mathbf{Y}_1, \ldots, \mathbf{Y}_N\}$ and $\mathbf{O} = \{\mathbf{O}_1, \ldots, \mathbf{O}_N\}$ denote these quantities for all time periods. Then the joint data likelihood model for $\mathbf{Y}$ can be re-written using the following first-order autoregressive decomposition

$$\begin{aligned} f(\mathbf{Y}_1, \ldots, \mathbf{Y}_N|\mathbf{O}) &= f(\mathbf{Y}_1|\mathbf{O}) \prod_{t=2}^{N} f(\mathbf{Y}_t|\mathbf{Y}_{t-1}, \ldots, \mathbf{Y}_1, \mathbf{O}) \\ &= f(\mathbf{Y}_1|\mathbf{O}_1) \prod_{t=2}^{N} f(\mathbf{Y}_t|\mathbf{O}_t, \mathbf{D}_{t-1}, \ldots, \mathbf{D}_1), \end{aligned} \tag{4}$$

where $\mathbf{D}_{t-j} = \{\mathbf{Y}_{t-j}, \mathbf{O}_{t-j}\}$ denotes the response and offset data for time period $t-j$. It is clear from (4) that with the exception of $\mathbf{Y}_1$, the distribution of $\mathbf{Y}_t$ can depend on the current offset as well as the set of past responses and offsets through $\mathbf{D}_{t-j}$. This insight provides two pointers for how to build a computationally efficient yet flexible spatio-temporal smoothing model. The first is that the spatio-temporal autocorrelations in the data at any time period $t > 1$ could be captured in a parsimonious manner using past data $\{\mathbf{D}_{t-1}, \ldots, \mathbf{D}_1\}$, rather than via highly parameterised and computationally demanding random effects structures. The second is that time period $t = 1$ is a special case because it has no past data to rely on, and hence will need a different model structure compared to times $t > 1$. Therefore, the proposed STARPOD model represents the structural component of the linear predictor $\boldsymbol{\xi}_t = \{\xi_t(\mathcal{A}_1), \ldots, \xi_t(\mathcal{A}_K)\}$ from (1) by

$$\boldsymbol{\xi}_t = \begin{cases} \boldsymbol{\phi} + \delta_t \mathbf{1} + \boldsymbol{\psi} & \text{if } t = 1 \\ \boldsymbol{\phi} + \delta_t \mathbf{1} + \sum_{j=1}^{\min\{p,t-1\}} \mathbf{v}_{t-j}(\mathbf{D}_{t-j})\alpha_{j_t} + \sum_{j=1}^{\min\{p,t-1\}} \mathbf{W}^* \mathbf{v}_{t-j}(\mathbf{D}_{t-j})\gamma_{j_t} & \text{if } t > 1, \end{cases} \tag{5}$$

where $\mathbf{1}$ denotes a $K \times 1$ vector of ones. Here, $\boldsymbol{\phi} = \{\phi(\mathcal{A}_1), \ldots, \phi(\mathcal{A}_K)\}$ is a vector of spatial random effects common to all time periods, which thus captures the average spatial pattern in disease rates for the entire study duration. Similarly, $\boldsymbol{\delta} = \{\delta_1, \ldots, \delta_N\}$ is a vector of temporal random effects common to all areal units, which thus captures the temporal trend for the entire study region. A number of different prior specifications are available for these random effects, and details are given in Section 2.2 below. The remaining terms in the linear predictor thus capture any deviations from these global spatial and temporal trends, and as there are no past data to use at time $t = 1$ these deviations are captured by a second set of spatial random effects $\boldsymbol{\psi} = \{\psi(\mathcal{A}_1), \ldots, \psi(\mathcal{A}_K)\}$. Thus, $\boldsymbol{\phi}$ represents the average spatial pattern in disease rates over all time periods, while $\boldsymbol{\psi}$ captures the differences between this time-averaged spatial pattern and the spatial pattern observed for time period $t = 1$. In initial model development, data were generated under the above model, and these two random effect components were identifiable from the data.

For any time period $t > 1$ the deviations from $(\boldsymbol{\phi}, \boldsymbol{\delta})$ are captured by a spatially smoothed observation-driven autoregressive process of order p, where $\mathbf{v}_r(\mathbf{D}_r) = \{v_r(\mathcal{A}_1, \mathbf{D}_r), \ldots, v_r(\mathcal{A}_K, \mathbf{D}_r)\}$ denotes a $K \times 1$ covariate vector constructed from the collection of past data $\mathbf{D}_r$ at time r. Note, time periods for which $r < 1$, i.e., $j \geq t$, would have missing (NA) values in $\mathbf{v}_r(\mathbf{D}_r)$ and are hence excluded from the sum in (5). The exact construction of $\mathbf{v}_r(\mathbf{D}_r)$ depends on the data likelihood model, and should approximate the spatial pattern in disease risks/rates on the linear predictor scale of the data. Thus, for a binomial logistic regression specification $v_r(\mathcal{A}_K, \mathbf{D}_r) = \ln\left\{[Y_r(\mathcal{A}_K)/O_r(\mathcal{A}_K)]/[1 - Y_r(\mathcal{A}_K)/O_r(\mathcal{A}_K)]\right\}$, while for a Poisson log-linear model $v_r(\mathcal{A}_K, \mathbf{D}_r) = \ln\left\{Y_r(\mathcal{A}_K)/O_r(\mathcal{A}_K)\right\}$. This construction models any local deviations from $(\boldsymbol{\phi}, \boldsymbol{\delta})$ with a regression component containing only a few unknown parameters, rather than by a complex set of random effects.

In the proposed model $\mathbf{W}^*$ is the row standardised transformation of the neighbourhood matrix $\mathbf{W}$ (the elements in each row sum to one), so that the final term in (5) for $t > 1$, i.e., $\mathbf{W}^* \mathbf{v}_r(\mathbf{D}_r)$, is the spatially weighted average of the covariate $\mathbf{v}_r(\mathbf{D}_r)$. Thus $\sum_{j=1}^{\min\{p,t-1\}} \mathbf{v}_{t-j}(\mathbf{D}_{t-j})\alpha_{j_t}$ captures any temporal autocorrelation in the data, because the value of the structural component $\xi_t(\mathcal{A}_k)$ depends explicitly on the past data values in $\mathcal{A}_k$ via $\{\mathbf{D}_{t-1}, \ldots, \mathbf{D}_{t-j}\}$. Similarly, spatio-temporal autocorrelation is captured via $\sum_{j=1}^{\min\{p,t-1\}} \mathbf{W}^* \mathbf{v}_{t-j}(\mathbf{D}_{t-j})\gamma_{j_t}$, which depends on local spatially weighted averages of the same past data. A time-varying coefficient prior is specified for each $\boldsymbol{\alpha}_j = (\alpha_{j_{j+1}}, \ldots, \alpha_{j_N})$ and $\boldsymbol{\gamma}_j = (\gamma_{j_{j+1}}, \ldots, \gamma_{j_N})$ for $j = 1, \ldots, p$, which allows the effects of the past data $\mathbf{v}_{t-j}(\mathbf{D}_{t-j})$ to induce varying levels of temporal dependence over time for maximum model flexibility. In initial simulation studies similar to those presented in Section 5 we also considered allowing these regression parameters to vary spatially, but as this did not improve the estimation performance and led to much slower model fitting this extension is not considered any further.

4.2. Level 2 - Random effects and hyperpriors

The two sets of spatial random effects $\{\boldsymbol{\phi}, \boldsymbol{\psi}\}$ are assigned the CAR prior proposed by Leroux et al. (2000), which has the multivariate Gaussian form

$$\begin{aligned} \boldsymbol{\phi} &\sim \mathrm{N}(\mathbf{0}, \tau_\phi\{\rho_\phi[\mathrm{diag}(\mathbf{W}\mathbf{1}) - \mathbf{W}] + (1 - \rho_\phi)\mathbf{I}\}^{-1}) \\ \boldsymbol{\psi} &\sim \mathrm{N}(\mathbf{0}, \tau_\psi\{\rho_\psi[\mathrm{diag}(\mathbf{W}\mathbf{1}) - \mathbf{W}] + (1 - \rho_\psi)\mathbf{I}\}^{-1}) \\ \tau_\phi^{-1/2}, \tau_\psi^{-1/2} &\sim \text{Half-Normal}(\tau_0 = 0.001) \\ \ln\left(\frac{\rho_\phi}{1 - \rho_\phi}\right), \ln\left(\frac{\rho_\psi}{1 - \rho_\psi}\right) &\sim \mathrm{N}(0, 0.25), \end{aligned} \tag{6}$$

where $(\mathbf{0}, \mathbf{1})$ are vectors of zeros and ones respectively, while $\mathbf{I}$ is the identity matrix. The prior precision parameters (τ_ϕ, τ_ψ) are assigned half-normal priors on the standard deviation scale (i.e., for $\tau_\phi^{-1/2}$ and $\tau_\psi^{-1/2}$) with precision parameter $\tau_0 = 0.001$ as suggested by Gelman (2006) to be weakly informative. The hyperparameters (ρ_ϕ, ρ_ψ) control the strength of the spatial dependence amongst the random effects, and are assigned zero-mean Gaussian priors with precision 0.25 on the logit scale to be weakly informative (see Section 2.1 of the supplementary material). As any temporal autocorrelation in the data should be captured by the covariates $\{\mathbf{v}_{t-1}(\mathbf{D}_{t-1}), \ldots, \mathbf{v}_{t-p}(\mathbf{D}_{t-p})\}$, we specify a computationally efficient independence prior for $\boldsymbol{\delta}$, namely

$$\begin{aligned} \boldsymbol{\delta} &\sim \mathrm{N}(\mathbf{0}, \tau_\delta^{-1}\mathbf{I}) \\ \ln(\tau_\delta^{-1/2}) &\sim \text{Half-Normal}(\tau_0 = 0.001). \end{aligned} \tag{7}$$

Temporally autocorrelated priors such as first-order autoregressive processes were considered when developing the model, but as they performed almost identically to (7) in initial simulation studies but were more computationally intensive to fit they are not considered here. Analogous independence shrinkage priors are assigned to the time-varying regression parameters $\{\boldsymbol{\alpha}_j, \boldsymbol{\gamma}_j\}$ for each $j = 1, \ldots, p$:

$$\begin{aligned} \boldsymbol{\alpha}_j &\sim \mathrm{N}(\mathbf{0}, \tau_{\alpha_j}^{-1}\mathbf{I}) \\ \boldsymbol{\gamma}_j &\sim \mathrm{N}(\mathbf{0}, \tau_{\gamma_j}^{-1}\mathbf{I}) \\ \ln(\tau_{\alpha_j}^{-1/2}), \ln(\tau_{\gamma_j}^{-1/2}) &\sim \text{Half-Normal}(\tau_0 = 0.001). \end{aligned} \tag{8}$$

More complex temporally autocorrelated priors were again considered when developing the model, but as they performed almost identically to (7) they are not considered further.

4.3. Model implementation

Software in the form of an R script file is provided in Section 2.2 of the supplementary material to allow others to fit the class of STARPOD models and the competitors outlined in this paper using the R package INLA. Additionally, for ease of reference, Section 2.3 of the supplementary material provides a summary of the models compared in this paper, including both their mathematical descriptions and their parametric complexity.

5. Simulation study

The aim of this study is to illustrate the general good performance of the STARPOD model proposed in Section 4, by comparing it to a number of competitor models commonly used in spatio-temporal disease mapping. We assess each models' estimation and predictive performance under a range of scenarios with different spatio-temporal disease rate patterns, which allows us to assess how the model performs across the board. This study mirrors the motivating application by focusing on binomial type prevalence data, and the results are presented below and in Section 3.2 of the supplementary material. Additionally, a second study examining Poisson type count data that are commonly used in disease mapping is presented in Section 3.3 of the supplementary material. Finally, Section 3.4 of the supplementary material presents a third small simulation study quantifying the relative computational times needed to fit the models compared in this paper for varying numbers of spatial units (K) and time periods (N).

5.1. Competitor models

Two variants of proposed STARPOD model are compared here, which have first (i.e., $p = 1$ and denoted STARPOD-1) and second (i.e., $p = 2$ and denoted STARPOD-2) order observation driven structures. Overdispersed versions of these models with beta-binomial rather than binomial data likelihoods were also considered, but as they produced almost identical results to the binomial models but were substantially slower to fit their results are not shown for brevity.

These models are compared against four commonly used competitors, which vary in their complexity, flexibility and computational overhead to fit. The first is the simple and computationally efficient separable model proposed by Knorr-Held and Besag (1998) and given by (2) without $\epsilon_t(\mathcal{A}_k)$, which is denoted here by SEP. The second and third are the non-separable models proposed by Knorr-Held (2000) and given by (2), where the second has independent interactions (denoted KN-I) while the third has correlated interactions in both time and space (denoted KN-IV). The fourth model is that proposed by Rushworth et al. (2014) and given by (3), which is denoted here by RUSH. Finally, we note that the divide and conquer approach to fitting model KN-IV is not considered here because its implementation via the R package bigDM only fits models with Poisson data likelihoods. However, this computationally efficient model fitting approach is compared in the Poisson-based simulation study presented in Section 3.2 of the supplementary material.

5.2. Data generation and scenarios

The study region is the set of $K = 4834$ LSOAs comprising Greater London, with the whole of England not used because the repeated model fitting, particularly KN-IV, would be computationally prohibitive for large numbers of simulated data sets. Data for this region are generated for $N = 10$ time periods, which again is smaller than that used in the motivating study for computational efficiency. Observed disease counts $\{Y_t(\mathcal{A}_k)\}$ for each simulated data set are generated from $Y_t(\mathcal{A}_k) \sim \text{Binomial}\{O_t(\mathcal{A}_k), \theta_t(\mathcal{A}_k)\}$, where the population sizes $\{O_t(\mathcal{A}_k)\}$ match those from the motivating application and the average prevalence is fixed to be the same as the motivating asthma data at just under 5%. One hundred data sets are generated under the following 5 different scenarios, which differ in the spatio-temporal structure of the true disease rate surface $\{\theta_t(\mathcal{A}_k)\}$.

- **Scenario 1 -** A spatially smooth rate surface whose mean level shows a decreasing linear trend over time but whose spatial pattern does not change. This mimics the situation of a static disease rate surface that is impacted by a population-wide intervention that has the same effect at reducing disease rates in all communities.
- **Scenario 2 -** A spatially smooth rate surface whose spatial pattern evolves slowly over time by depending on the rate surface from the previous time period. This mimics the spatio-temporal dynamics for a chronic disease where each area has a different slow evolution in rate over time due to a range of local factors.
- **Scenario 3 -** A spatially smooth rate surface whose spatial pattern becomes more variable over time. This mimics a chronic disease where there is an increase in the spatial inequality in rates over time, i.e., the high-rate areas exhibit increasing rates over time while the low-rate areas exhibit decreasing rates.
- **Scenario 4 -** A spatially smooth rate surface whose spatial pattern evolves non-linearly over time due to the influence of both within-area effects and spatial spillovers at time $t-1$.
- **Scenario 5 -** A locally spatially smooth rate surface, whose spatial pattern evolves slowly and non-linearly over time. This mimics a chronic disease that exhibits the same subset of clusters of areas with elevated (or reduced) rates compared to their neighbouring areas for all time periods.

Mathematical descriptions of the data generation for each of these five scenarios are given in Section 3.1 of the supplementary material.

5.3. Measuring model performance

A random selection of 4% (1934 out of 48,340) of the data are treated as missing values when fitting the models, which allows us to assess the spatial predictive ability of the models as well as their ability to estimate the true disease rates in areas where there are noisy data values. The following metrics are used to assess model performance: (i) `computational time` - measured by the time taken in seconds to fit each model; (ii) `accuracy of disease rate estimation` - measured by the root mean square error (RMSE) and the median absolute error (MAE) of the estimated (posterior mean) disease rates (prevalences) $\{\theta_t(\mathcal{A}_k)\}$; and (iii) `rate uncertainty quantification` - measured by the coverage probabilities (CP) of the 95% credible intervals for the true disease rates $\{\theta_t(\mathcal{A}_k)\}$ and the mean widths of these intervals (MIW). Both RMSE and MAE are presented because the latter is robust to outlying estimates while the former is adversely affected by them due to its squared and mean operators. Note, we do not present the metrics in (ii) and (iii) for individual model parameters because: (i) the data from the 5 scenarios were generated under different models and hence different parameterisations; and (ii) the different competitor models also have different parameterisations, and hence the parameters are not consistent across the different models.

The results of (ii) and (iii) are averaged over all 100 replicated data sets and all spatio-temporal units, except where they are separated by in-sample rate estimation (96% of data points) and out-of-sample rate prediction (4% of data points). The simulation study was run on compute servers due to their computational convenience, but this would not give users a realistic summary of the time taken to fit each model on their own desktop computer. Therefore, a smaller study with 10 data sets from each scenario was run on an iMac desktop computer with a 3.8 GHz 8-Core Intel Core i7 processor and 32 GB of memory, and the computational time results presented below are averaged over this smaller set.

5.4. Results

The results of the computational speed and in-sample prevalence estimation are presented in Table 1, while the out-of-sample predictive performance results are summarised in Section 3.2 of the supplementary material. In scenario 1 the disease rate structure is separable in space and time, and in this simplistic situation most models perform similarly, with the separable model (SEP) being preferred due to its vastly superior computational speed. However, in the remaining scenarios with more realistic spatio-temporal disease rate structures the separable model performs very poorly, having the highest RMSE, the highest or second highest MAE, and very narrow 95% credible intervals that are not near the required nominal coverage levels (between 59% and 84%).

Leaving aside the separable model, both STARPOD models are much quicker to implement than the remaining competitors in all cases, with STARPOD-1 naturally requiring less computational time than STARPOD-2. For example, the first-order STARPOD model only takes between 17% and 35% of the computational time required by the Rushworth model (RUSH), while the corresponding comparisons with KN-I are 15% to 46% and with KN-IV are 6% to 15%. This large improvement in computational efficiency does not come at the cost of vastly reduced estimation performance, as both STARPOD-1 and STARPOD-2 have RMSE and MAE values that are similar to those from the three non-separable competitors. For example, both STARPOD models have either the same or lower RMSE and MAE values than the KN-I and KN-IV models in four of the five scenarios, and only slightly higher values than KN-IV in scenario 5 (by 0.0002). Even compared to the RUSH model the estimation performance is largely comparable, with the largest differences in RMSE and MAE being 0.0004 (RMSE) and 0.0002 (MAE) in scenarios 2 and 5.

The only facet where the STARPOD models perform markedly less well than their competitors is in uncertainty quantification, as their coverage probabilities are slightly too low and range between 85.1% and 96.2% depending on the scenario. This contrasts to the KN-IV and RUSH models whose coverage probabilities range between 95.1% and 97.7%. However, the reduced coverages from the STARPOD models are accompanied by noticeably narrower credible intervals compared to those from the KN-I, KN-IV and RUSH models. This suggests that when quantifying disease rate uncertainty from the STARPOD models it might be prudent to take a slightly elevated credible interval level, which should have similar widths to the 95% intervals from the competitor models.

Table 1
Results of the simulation study comparing the speed and in-sample estimation performance of the 6 models outlined in Section 5.1. Here, time is measured in seconds, while the remaining metrics denote: RMSE — root mean square error; MAE — median absolute error; CP — coverage probability; MIW — mean interval width. The best value in each row is in bold for clarity, except for MIW where the value is dependent on the coverage probability.

Scenario	Model					
	SEP	KN-I	KN-IV	RUSH	STARPOD-1	STARPOD-2
Time						
1	**17.3**	156.5	489.1	210.3	72.7	142.4
2	**16.4**	200.9	863.1	237.0	55.7	96.9
3	**22.2**	137.0	689.0	284.8	48.8	93.2
4	**16.9**	342.1	719.6	280.6	51.2	97.8
5	**25.1**	213.5	899.8	313.2	62.8	124.0
RMSE						
1	**0.0012**	**0.0012**	0.0016	**0.0012**	**0.0012**	**0.0012**
2	0.0019	0.0019	0.0017	**0.0012**	0.0016	0.0016
3	0.0063	0.0036	0.0021	**0.0020**	0.0021	**0.0020**
4	0.0031	0.0028	0.0020	0.0017	**0.0016**	**0.0016**
5	0.0023	0.0022	0.0018	**0.0017**	0.0020	0.0020
MAE						
1	**0.0008**	**0.0008**	0.0011	**0.0008**	**0.0008**	**0.0008**
2	0.0012	0.0012	0.0012	**0.0008**	0.0010	0.0010
3	0.0014	0.0015	0.0013	0.0011	**0.0010**	**0.0010**
4	0.0014	0.0015	0.0013	0.0011	**0.0010**	**0.0010**
5	0.0014	0.0014	**0.0011**	**0.0011**	0.0013	0.0013
CP						
1	0.962	0.964	**0.951**	0.964	0.962	0.963
2	0.784	0.888	**0.963**	0.977	0.851	0.851
3	0.590	**0.936**	0.970	0.971	0.878	0.884
4	0.750	0.894	**0.962**	0.970	0.927	0.931
5	0.840	0.923	**0.963**	0.964	0.889	0.892
MIW						
1	0.0049	0.0049	0.0064	0.0050	0.0049	0.0049
2	0.0046	0.0062	0.0073	0.0057	0.0047	0.0046
3	0.0059	0.0133	0.0089	0.0085	0.0060	0.0058
4	0.0053	0.0090	0.0081	0.0073	0.0057	0.0057
5	0.0061	0.0079	0.0073	0.0071	0.0065	0.0065

6. Results of the England study

6.1. Model comparison

The six models compared in the simulation study were fitted to the four disease data sets from the motivating study outlined in Section 2, and a comparison of their relative performances is presented in Table 2. The table presents a number of performance metrics, including the time taken in seconds to fit each model, and a number of model fit and prediction metrics. The model fit metrics include the deviance information criterion (DIC, Spiegelhalter et al., 2002) that one wishes to minimise, the log marginal predictive likelihood, (LMPL, Geisser, 1980) that one needs to maximise, and the average level of spatial autocorrelation in the model residuals as measured by the mean Moran's I (Moran, 1950) statistic over the 13-year study period. Finally, we also measure each model's out-of-sample predictive performance, by computing the median absolute error of its predictions when splitting the disease data at random into a 95% training set and a 5% test set (21,288 observations).

The table shows some conflicting results across the metrics, although these differences are fairly consistent across the four diseases. The separable (SEP) and the type I interaction (KN-I) models perform similarly in most cases, which is because for these data the variance of the independent spatio-temporal interactions is almost zero and hence the two models are almost identical. Thus, as KN-I takes longer to fit than SEP but offers no improvement in performance, the former is not considered further. The separable model is the fastest to fit in all cases, while the two STARPOD models are much faster than the non-separable KN-IV and RUSH models. Both STARPOD models also exhibit the best fit to the observed data and the best predictive performance out of all candidate models, having better DIC (lower), LMPL (higher) and MAE (lower) values than the more complex KN-IV and RUSH models. For most of these metrics the STARPOD-2 model is superior to the STARPOD-1 model, with the exception of MAE where this trend is reversed. None of the candidate models capture all of the spatial autocorrelation in the data, as the time-averaged Moran's I statistics are all greater than zero. They do capture the majority of this correlation however, as the corresponding Moran's I values for the raw prevalence data are: Asthma - 0.496; Cancer - 0.476; CHD - 0.632; COPD - 0.526. The Rushworth model performs the best at capturing this spatial autocorrelation, while the two STARPOD models are in the middle of the candidate models in this regard. Thus, as the two STARPOD models exhibit the best fit to the observed data and the best predictive performance as illustrated above, the remainder of the results presented in this section come from this model class. Specifically, we use the STARPOD-2 model

Table 2
Comparison of the relative performance of each model, including their fitting time (in seconds), their overall model fit via the deviance information criterion (DIC) and the log marginal predictive likelihood (LMPL), the average level of unexplained spatial autocorrelation (Moran's I) and their out-of-sample predictive ability for the disease counts (MAE). The best value in each column for each disease outcome is in bold for clarity.

Model	Time	DIC (p.d)	LMPL	Moran's I	MAE
Asthma					
Sep	**173**	2,857,144 (30,171)	−1,418,111	0.191	2.70
KN-I	877	2,857,148 (30,171)	−1,418,113	0.191	2.70
KN-IV	20,601	2,847,341 (39,952)	−1,408,852	0.045	2.28
Rush	3588	2,845,739 (37,563)	−1,409,006	**0.031**	2.35
STARPOD-1	1303	2,791,546 (10,750)	−1,397,101	0.143	**1.93**
STARPOD-2	1960	**2,790,369 (10,727)**	**−1,396,531**	0.146	2.05
Cancer					
Sep	**174**	2,236,121 (30,564)	−1,106,303	0.198	1.15
KN-I	904	2,236,124 (30,568)	−1,106,302	0.198	1.15
KN-IV	6790	2,232,069 (36,187)	−1,102,283	0.090	1.00
Rush	9911	2,230,691 (33,827)	−1,101,932	**0.069**	1.05
STARPOD-1	1028	2,168,245 (12,299)	−1,083,569	0.131	**0.91**
STARPOD-2	1648	**2,167,505 (12,332)**	**−1,083,192**	0.133	0.96
CHD					
Sep	**233**	2,582,289 (30,846)	−1,278,986	0.245	1.52
KN-I	801	2,582,289 (30,846)	−1,278,986	0.245	1.52
KN-IV	11,299	2,574,586 (35,911)	−1,273,404	0.092	1.29
Rush	6209	2,574,104 (34,565)	−1,272,908	**0.073**	1.33
STARPOD-1	1484	2,537,971 (16,188)	−1,272,542	0.127	**0.97**
STARPOD-2	1983	**2,535,416 (15,938)**	**−1,271,420**	0.142	1.09
COPD					
Sep	**183**	2,295,104 (31,171)	−1,136,175	0.209	1.36
KN-I	1241	2,295,107 (31,174)	−1,136,175	0.209	1.36
KN-IV	8224	2,283,568 (40,312)	−1,126,876	0.039	1.13
Rush	5367	2,282,595 (38,642)	−1,126,308	**0.027**	1.17
STARPOD-1	1360	2,220,984 (14,233)	−1,114,313	0.123	**0.89**
STARPOD-2	1586	**2,220,650 (14,210)**	**−1,114,173**	0.123	0.98

for all four diseases due to its superior in-sample fit to the data. Motivating questions (A) and (B) from Section 2 are addressed in what follows, while question (C) is answered in Section 4 of the supplementary material.

6.2. (A) Temporal trends and regional differences

The temporal trends in the estimated average prevalences (posterior mean) for each of the 9 regions in England are shown in Fig. 2, where the four panels relate to: (A) Asthma; (B) Cancer; (C) Coronary heart Disease; and (D) Chronic Obstructive Pulmonary Disease. In each panel the black dashed line depicts the England-wide national average. The figure shows there is a marked similarity between the 9 regional temporal trends for each disease, as they exhibit almost the same trajectory in all cases. Focusing on the trends over the entire country (black dashed lines) shows that asthma prevalence exhibits almost no change over time, as the average prevalence has increased by 0.1% in 2017 compared to 2005 (a prevalence of 6.0% compared to 5.9%). In contrast, cancer and COPD exhibit increasing prevalences by 2.1% (cancer) and 0.5% (COPD) over the 13-year period, while CHD exhibits a decreasing prevalence of around 0.4%. There are some regional disparities apparent from the figure, with London having the lowest prevalences for all four diseases for all years. The disparity between London and the rest of the country is particularly marked for asthma and cancer, because of the high degree of similarity between the remaining 8 regions. The highest average disease prevalences are found in the South West for Asthma and Cancer and in the North East for CHD and COPD, which is again consistent over all time periods.

6.3. (B) Health inequalities

The levels of total inequality in disease prevalence within each of the 9 regions of England are shown in Fig. 3, where the four panels again relate to the four diseases. Within each year and region total inequality is measured by the standard deviation in the posterior mean prevalences between LSOAs within that region for that year. For comparison, the black dashed lines show the total inequality across England, which is measured by the standard deviation in the posterior mean prevalences between all LSOAs in England. The figure shows that total inequality has increased for asthma, cancer and COPD between 2005 and 2017, while for CHD it increased from 2005 to 2006 before exhibiting a decreasing trend thereafter. The biggest increase in inequality is observed for cancer prevalence, which across England had a between LSOA standard deviation of 0.076 in 2005 which rose 10-fold to 0.77 in 2017. The 9 regions show markedly similar inequality trends for each disease, with the highest inequality being observed in the East of England for asthma, in the South West for cancer, in the North West for CHD and in the North East for COPD. In contrast,

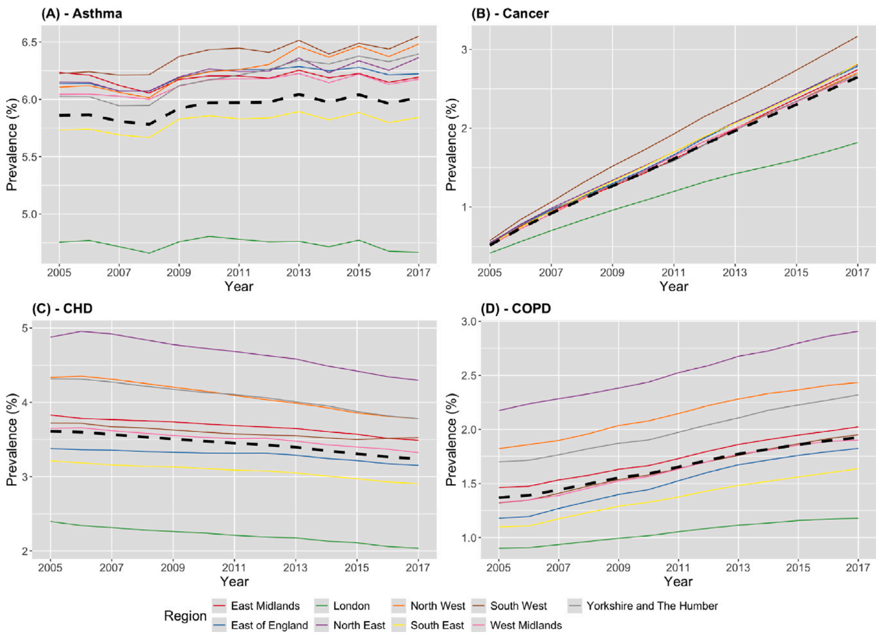

Fig. 2. Temporal trends in the average posterior mean prevalence estimates from the STARPOD-2 model for each of the 9 regions in England: (A) Asthma; (B) Cancer; (C) Coronary Heart Disease; and (D) Chronic Obstructive Pulmonary Disease. In each panel the black dashed line shows the England wide average.

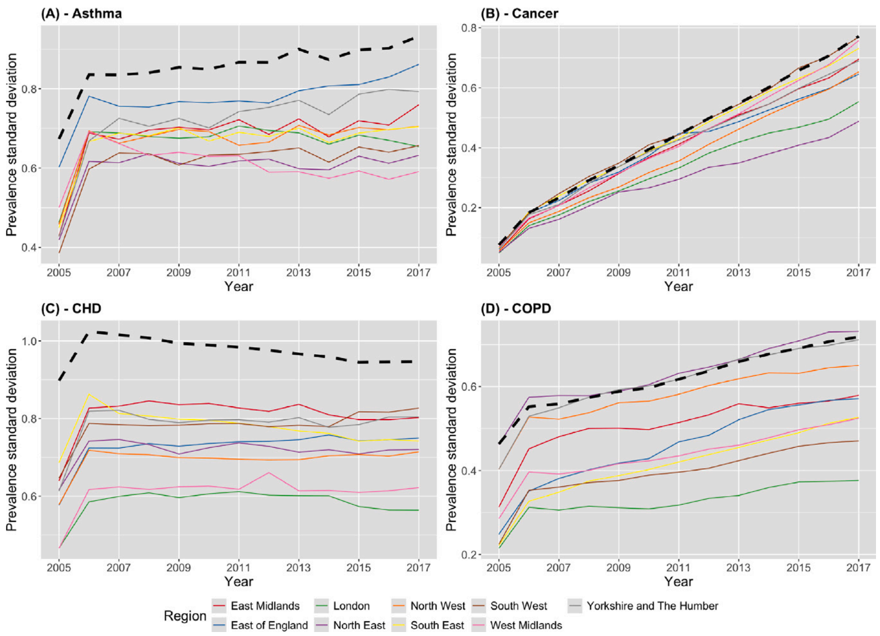

Fig. 3. Temporal trends in the levels of total inequality within each of the 9 regions in England: (A) Asthma; (B) Cancer; (C) Coronary Heart Disease; and (D) Chronic Obstructive Pulmonary Disease. Total inequality is measured by the standard deviation of the LSOA-level posterior mean prevalences estimated from the STARPOD-2 model. In each panel the black dashed line shows the total inequality across England.

the lowest levels of total inequality are observed in London for CHD/COPD and in the North East for cancer, while there is no clear pattern with respect to asthma.

The levels of socio-economic inequality in disease prevalence are assessed using deciles of the English index of multiple deprivation (IMD) from 2015, with decile 1 being the most deprived while decile 10 is the most affluent. Fig. 4 presents the levels of socio-economic inequality for each of the 9 regions of England, by displaying the mean estimated prevalences over all

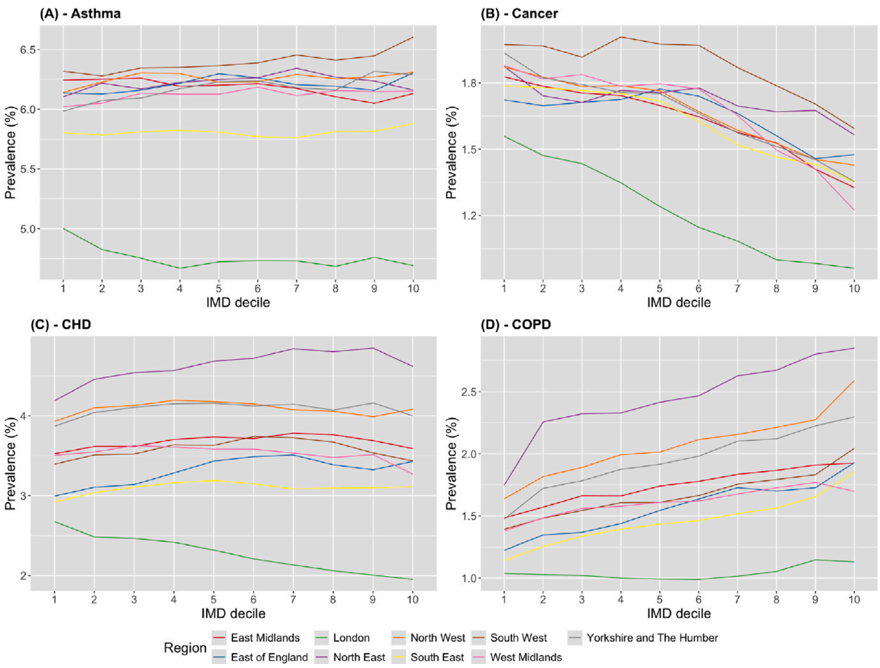

Fig. 4. Socio-economic inequality in the estimated mean prevalence from the STARPOD-2 model within each of the 9 regions in England: (A) Asthma; (B) Cancer; (C) Coronary Heart Disease; and (D) Chronic Obstructive Pulmonary Disease. In each panel decile 1 is the most deprived while decile 10 is the most affluent.

13 years of the study for each IMD decile, region and disease. Exploratory analyses also examined the variation in socio-economic deprivation over time, but as the results showed no major temporal trends they are not shown for brevity. The figure shows that for asthma and CHD there are no clear socio-economic inequalities in disease prevalence for most regions, with the exception being in London where for example the average prevalence for CHD is 2.68% for the most deprived decile which drops to 1.95% for the most affluent one. In contrast, cancer exhibits the most consistent socio-economic inequalities, with the most deprived communities having higher prevalences than more affluent ones for all regions. For example, in the South West the average prevalence ranges between 1.97% (most deprived) to 1.59% (most affluent), while the corresponding results for London are 1.56% to 0.96%. For COPD the socio-economic inequality gradient is generally reversed (except for London), with higher prevalences being observed in more affluent communities than in deprived ones. For example, in the North East the mean prevalence is 1.75% for the most deprived decile, which rises to 2.85% for the most affluent one.

7. Discussion

This paper has proposed a computationally efficient spatio-temporal disease mapping model called STARPOD, which is primarily intended for analysing big data sets. The model combines the computationally efficient but restrictive separable spatio-temporal structure proposed by Knorr-Held and Besag (1998) with a spatially smoothed observation-driven component, the latter of which was motivated by the time series models originally developed by Zeger and Qaqish (1988). The simulation and real data studies have shown that under a variety of spatio-temporal disease rate structures this model exhibits close to the estimation and prediction performance of the highly parameterised models proposed by Knorr-Held (2000) and Rushworth et al. (2014), which are commonly used in disease mapping applications. The key advantage of our contribution is its computational efficiency, because it can achieve this close to comparable estimation performance while only taking between 5% to 40% of the time to implement compared to the above more complex models. The STARPOD model class is straightforward to implement using the `R` package `INLA`, and example code is provided with this paper to allow others to apply it to their own data.

The simulation studies have shown that the vast computational improvements offered by the STARPOD model do not come with a large loss of estimation and prediction accuracy, with most of the scenarios (except scenario 2) showing that the model performs as well as that proposed by Rushworth et al. (2014) and slightly better than that proposed by Knorr-Held (2000) with type IV interactions. However, this improved computational speed does come at the cost of a slight reduction in the appropriateness of the 95% credible intervals for the STARPOD models, with coverage probabilities typically between 85% and 95%. This reduction is likely to be because the model only contains simple separable random effects structures, which do not induce as much posterior uncertainty in the disease rates compared to the non-separable random effect structures utilised by the more complex models. However, these reduced coverages are accompanied by narrower credible intervals, so if uncertainty quantification is a key focus then we recommend taking an elevated credible interval level, which should have similar widths to the 95% intervals from the competitor models.

The other limitation of the STARPOD models is over the inclusion of covariates, which was not considered in this paper due to a lack of available covariate data for the motivating application. However, initial simulation studies were run with covariates, and the STARPOD models performed worse than the commonly used competitors mentioned above in this context. The reason for this is that in the presence of covariates the construction of $\{\mathbf{v}_r(\mathbf{D}_r)\}$ has to be adjusted by estimating initial covariate effects based on only $\mathbf{D}_r$, and the additional error and uncertainty in these covariate effect estimates propagates through to produce errors in $\{\mathbf{v}_r(\mathbf{D}_r)\}$. As a result, the observation-driven component of the STARPOD models do not capture the residual spatial structure in the data as well, resulting in poorer model performance. One possible solution to investigate in future work is to use measurement error models, which would allow for these errors in the observation-driven covariates within the STARPOD models. Thus, in their current form we do not recommend using the STARPOD models for spatial regression applications, such as quantifying the effects of air pollution on human health (Lee et al., 2022), identifying the factors influencing property prices (Kawabata et al., 2022), or assessing regional personality traits (Ebert et al., 2022). Whilst the set of ecological regression applications is large so to is the field of spatio-temporal disease mapping, which gives the STARPOD models a large user-base, especially as researchers gain access to ever bigger and more complex data sets.

Finally, the motivating study has shown that the prevalences of cancer and COPD have increased over the 13-year study period, while the prevalence of CHD has declined and the prevalence of asthma has remained largely unchanged. Additionally, each disease exhibits sizeable regional inequality in the estimated prevalences, with London having the lowest average rates while the North East and the South West have the largest prevalences depending on the disease. Finally, the prevalences exhibit some level of socio-economic inequality, with cancer exhibiting increased rates for poorer communities while the opposite pattern is true for COPD. However, when interpreting these results one must remember that they relate to unadjusted prevalences, and thus give insight into the overall amount of health care services needed by communities with different levels of deprivation. However, they do not factor in the varying age structures of these deprived and affluent communities, and thus should not be interpreted in terms of being the differential *risks* of disease for different socio-economic groups. For example, more deprived communities are younger on average than more affluent ones (see for example Pratt, 2021), and hence any disease that has a very large age gradient in its prevalence may have a higher overall prevalence in more affluent communities compared to more deprived ones due to their differential age profiles. This phenomenon is the most likely cause of the socio-economic age gradient observed for COPD, because for example the Global Burden of Disease study (GBD, https://vizhub.healthdata.org/gbd-results) estimates that in 2015 COPD prevalences in England were 15.8% in people aged 65 to 74 and 30.2% in people aged 75 to 84.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.spasta.2025.100901.

Data availability

The publicly available data are provided by the Place-Based Longitudinal Data Resource (PLDR) at https://pldr.org/. Software to implement the models compared in this paper on a simulated data set using the R package INLA is provided at the Github repository https://github.com/duncanplee/Big-data-disease-mapping.

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